LEVEL 10 — Conditional Operations

Very useful in Data Analytics — line-by-line examples with outputs

First line

```
import numpy as np
```

1. Boolean indexing

Select values using a True/False condition.

```
arr=np.array([10,20,30,40,50])
result=arr[arr>25]
print(result)
```

Output: `[30 40 50]`

2. Boolean arrays

A comparison creates an array of True and False.

```
arr=np.array([10,20,30,40])
print(arr>25)
```

Output: `[False False True True]`

3. Comparison operators

Use >, <, >=, <=, ==, and !=.

```
arr=np.array([10,20,30])
print(arr>=20)
```

Output: `[False True True]`

4. Logical conditions

Use & for AND and | for OR. Put conditions in parentheses.

```
arr=np.array([10,20,30,40,50])
result=arr[(arr>20)&(arr<50)]
print(result)
```

Output: `[30 40]`

5. np.where()

Works like if-else when given condition, true value, and false value.

```
marks=np.array([35,75,45,90])
result=np.where(marks>=50,'Pass','Fail')
print(result)
```

Output: `['Fail' 'Pass' 'Fail' 'Pass']`

6. np.select()

Handles multiple conditions.

```
marks=np.array([35,55,75,90])
conditions=[marks<40,marks<60,marks>=60]
choices=['Fail','Average','Good']
result=np.select(conditions,choices)
print(result)
```

Output: `['Fail' 'Average' 'Good' 'Good']`

7. np.any()

Returns True if at least one element satisfies the condition.

```
arr=np.array([10,20,30])
print(np.any(arr>25))
```

Output: `True`

8. np.all()

Returns True only if every element satisfies the condition.

```
arr=np.array([10,20,30])
print(np.all(arr>5))
```

Output: `True`

9. np.isin()

Checks whether each value exists in another collection.

```
arr=np.array([10,20,30,40])
result=np.isin(arr,[20,40])
print(result)
```

Output: `[False True False True]`

10. Data Analytics example

Filter passing students.

```
marks=np.array([35,67,82,45,90])
passed=marks[marks>=50]
print(passed)
```

Output: `[67 82 90]`

Quick Revision

`Boolean indexing` → filter values
`Boolean array` → True/False array
`> <, >=, <=, ==, !=` → comparisons
& → AND | | → OR
`np.where()` → if-else selection
`np.select()` → multiple conditions
`np.any()` → at least one True
`np.all()` → all True
`np.isin()` → membership check